

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.add("GraphRecipes")
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Resolving package versions...
No Changes to `C:\Users\sarah\OneDrive\Documents\GitHub\hw2-sarah-nathan\Project.toml`
No Changes to `C:\Users\sarah\OneDrive\Documents\GitHub\hw2-sarah-nathan\Manifest.toml`
Activating project at `c:\Users\sarah\OneDrive\Documents\GitHub\hw2-sarah-nathan`
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
using GraphRecipes
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

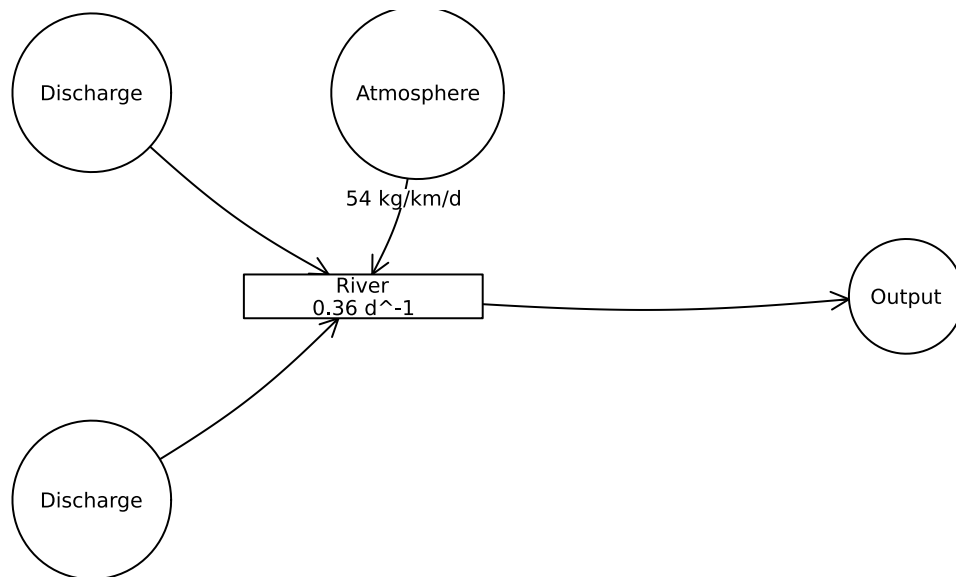
Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

```

A = [0 0 0 1 0;
     0 0 0 1 0;
     0 0 0 1 0;
     0 0 0 0 1;
     0 0 0 0 0]

graphplot(
    A,
    names = ["Discharge", "Discharge", "Atmosphere", "River\n0.36 d^-1", "Output"],
    nodeshape = [:circle, :circle, :circle, :rect, :circle],
    edgelabel = Dict{(3, 4) => "54 kg/km/d"},
    nodecolor = :white,
    markersize = 0.2,
    x = [-1.0, -1.0, 0.2, 0.0, 2.0],
    y = [0.75, -0.75, 0.75, 0.0, 0.0],
    arrow = true
)

```



Differential equation for change in CRUD concentration

$$\frac{dC}{dx} = \frac{a}{Q} - \frac{kC}{v}$$

Variables defined in cell below.

```
# Parameters
v = 10.0          # river velocity (km/d)
k = 0.36          # decay rate (d^-1)
a = 54.0          # atmospheric deposition (kg/km/d)
Q = 250000        # river flow rate (m3/d)
Q_d = 0.5 / 1000  # CRUD concentration in river inflow (kg/m3)
s1 = 40000        # discharge source 1 flow rate (m3/d)
s1_d = 9.0 / 1000 # CRUD concentration in discharge source 1 (kg/m3)
s2 = 60000        # discharge source 2 flow rate (m3/d)
s2_d = 7.0 / 1000 # CRUD concentration in discharge source 2 (kg/m3)

x1 = 0            # position of source 1 (km)
x2 = 15           # position of source 2 (km)

# Equilibrium concentration
function no_mix(Q)
    atmosphere = a / (k*Q)
    return atmosphere
end

# Mixing at each stopping point
function mix(Q_i, C_i, dis_flow, dis_conc)
    Q_f = Q_i + dis_flow
    C_f = (Q_i*C_i + dis_flow*dis_conc) / Q_f
    return Q_f, C_f
end

# Initial concentration
C_0 = Q_d

# Initial to discharge 1
```

0.0005